

Problem

The responses of a series RLC circuit are

$$v_c(t) = [30 - 10e^{-20t} + 30e^{-10t}]u(t) \text{ V}$$

$$i_L(t) = [40e^{-20t} - 60e^{-10t}]u(t) \text{ mA}$$

where $v_C(t)$ and $i_L(t)$ are the capacitor voltage and inductor current, respectively. Determine the values of R , L , and C .

Step-by-step solution

Step 1 of 8

The series RLC circuit is shown in Figure 1.

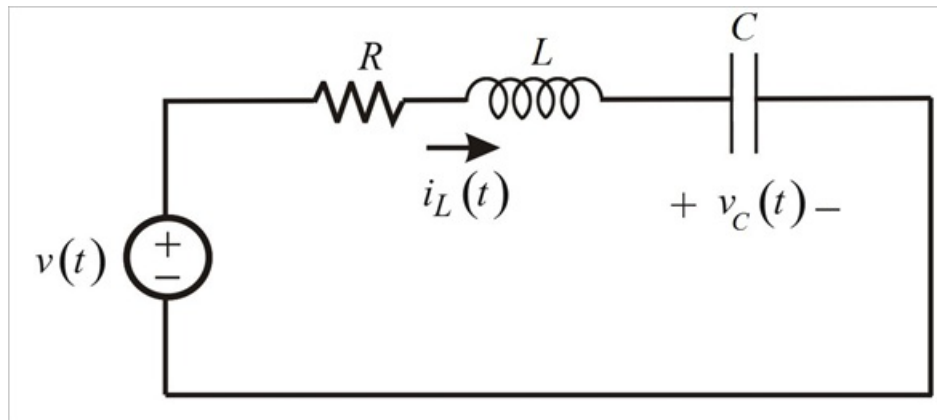


Figure 1

Step 2 of 8

The current through the inductor and the voltage across the capacitor are,

$$i_L(t) = (40e^{-20t} - 60e^{-10t})u(t) \text{ mA} \dots\dots (1)$$

$$v_C(t) = (30 - 10e^{-20t} + 30e^{-10t})u(t) \text{ V} \dots\dots (2)$$

The capacitor current and capacitor voltage are related as,

$$i_C(t) = C \frac{dv_C(t)}{dt}$$

In series circuit, the current is same through every element.

$$i_C(t) = i_L(t) \dots\dots (3)$$

Step 3 of 8

Substitute $C \frac{dv_c(t)}{dt}$ for $i_c(t)$ in equation (3).

$$i_L(t) = C \frac{dv_c(t)}{dt} \dots\dots (4)$$

Substitute equation (1) and (2) in equation (4).

$$(40e^{-20t} - 60e^{-10t}) \times 10^{-3} = C \frac{d}{dt} (30 - 10e^{-20t} + 30e^{-10t})$$

$$(40e^{-20t} - 60e^{-10t}) \times 10^{-3} = C (200e^{-20t} - 300e^{-10t}) \dots\dots (5)$$

Compare the exponential terms in equation (5).

$$40 \times 10^{-3} = 200C$$

$$C = 200 \times 10^{-6}$$

$$C = 200 \mu\text{F}$$

Step 4 of 8

The series RLC circuit in frequency domain is shown in Figure 2,

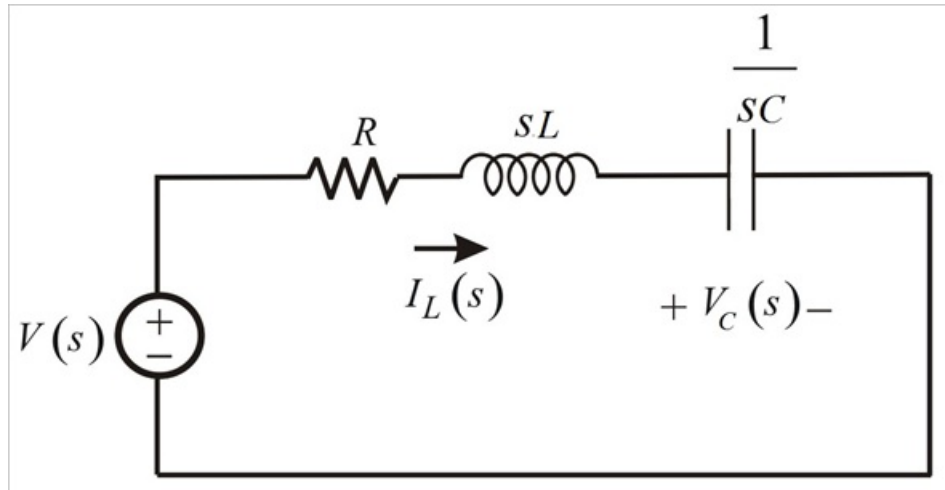


Figure 2

Step 5 of 8

The inductor current is,

$$i_L(t) = (40e^{-20t} - 60e^{-10t})u(t) \text{ mA}$$

Apply Laplace transform.

$$I_L(s) = \left(\frac{40}{s+20} - \frac{60}{s+10} \right) (10^{-3})$$

$$= \left[\frac{40(s+10) - 60(s+20)}{(s+10)(s+20)} \right] (10^{-3})$$

$$I_L(s) = \left(\frac{-20s - 80}{s^2 + 30s + 200} \right) (10^{-3}) \dots\dots (6)$$

Step 6 of 8

The impedance of the circuit in Figure 2.

$$Z(s) = R + sL + \frac{1}{sC}$$

$$Z(s) = \frac{sRC + s^2LC + 1}{sC} \dots\dots (7)$$

The current, voltage and impedance in the circuit in Figure 2 are related as,

$$\begin{aligned} I_L(s) &= \frac{V(s)}{Z(s)} \\ &= \frac{V(s)}{\frac{sRC + s^2LC + 1}{sC}} \\ &= \frac{sC}{sRC + s^2LC + 1} V(s) \end{aligned}$$

$$I_L(s) = \frac{1}{L} \left(\frac{s}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \right) V(s) \dots\dots (8)$$

Step 7 of 8

Compare denominators of equation (6) and (8).

$$s^2 + 30s + 200 = s^2 + \frac{R}{L}s + \frac{1}{LC} \dots\dots (9)$$

Compare the constant terms in equation (9).

$$\frac{1}{LC} = 200$$

Substitute $200 \mu\text{F}$ for C .

$$\begin{aligned} L &= \frac{1}{(200)(200 \times 10^{-6})} \\ &= \frac{1}{400 \times 10^{-2}} \\ &= 25 \text{ H} \end{aligned}$$

Step 8 of 8

Compare the coefficient of s in the equation (9) and substitute 25 H for L .

$$\begin{aligned} \frac{R}{L} &= 30 \\ R &= (L)(30) \\ &= (25)(30) \\ &= 750 \Omega \end{aligned}$$

Therefore, the values of R , L and C are 750Ω , 25 H and $200 \mu\text{F}$.